ChE 425 Nanotechnology for Pharmaceutical Applications

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Course format: synchronized zoom meetings for the first two weeks and likely to be in-person since week 3.

Lecture time: Tuesday 1 – 3:30 pm

Location: EIB RM 271

Office hour: Friday 1-3pm or other time by appointment

Textbook and reading materials:
Reading materials from high-impact journals and a few textbooks, including but not limited to,
Drug Delivery: Principles and Applications. Binghe Wang, Longqin He, Teruna J. Siahaan, Second edition, Wiley.
Drug Delivery Strategies for Poorly Water-soluble Drugs. Dennis Douroumis and Alfred Fahr, Wiley.
Physical Chemistry of Surfaces, by Arthur W. Adamson and Alice P. Gast, 6th edition
Science, Nature, JACS, PRL, Biomacromolecules, ACS Nano, Nano Letters, Soft Matter…

1. Topical outline

Basic concepts of nanoparticles for drug delivery
- Challenges facing the pharmaceutical industry
- Advantages of using nanoparticles
- Targeting strategies

Soft-matter nanoparticulate systems in therapeutics
- Lipidic nanosystems (Lipidic drug carriers, Liposomes...)
- Polymeric nanosystems (Polymerosomes, Dendrimers…)
- Mixed nanosystems (Hybridic nanosystems, Chimeric nanosystems…)

Self-assembled delivery vehicles: basic theoretical considerations and modeling concepts
- Brief reminder of equilibrium thermodynamics
- Principles of self-assembly in dilute solutions
- Solubility and partitioning of drugs
- Micelles and micellization
- Physicochemical properties of nanoparticles
- Drug loading

Nanoparticle characterization
- Dynamic light scattering
- X-ray scattering
- Neutron scattering

Stability of nanosuspensions
- Crystal growth and nucleation theory

- Creating supersaturation and stable nanosuspensions
- Antisolvent precipitation via mixer processing
- Particle engineering by spray
- Nanoparticle resuspension by using ultrasonication

Liposomes as intravenous solubilizers
- Solubilizing vehicles with precipitation risk upon dilution
- Solubilizing vehicles maintaining solubilization capacity upon dilution
- Mechanistic release aspects

Polymeric nanoparticle drug delivery systems
- Safety and biocompatibility of polymers
- Biodistribution and toxicity
- Encapsulation techniques
- Drug release and kinetics of drug transfer from mobile nanocarriers

Strategies for controlled drug delivery
- Thermo-responsive release
- Enzyme responsive release
- pH responsive release

Delivery of genes and oligonucleotides.
- Systemic delivery
- Cellular delivery
- Current and future approaches

Delivery of therapeutic cells
- T-cell delivery
- Islet cell encapsulation and implant

Nanosimilars and Regulatory aspects
- Nanosimilars
- Regulatory issues of nanomedicines

2. Weekly reading assignment and journal club presentation

There will be one journal club presentations at the beginning of every lecture, which summarizes the reading assignment of the week. The person, who is presenting the journal, is randomly selected. The journal club presentation is about 15 – 20 minutes.

3. Distribution of final grade

Class attendance and discussion: 20%
- Active class discussion
- Class attendance

Journal presentation: 20%
Mid-term review: 30%

Final presentation: 30%

4. Course policies

Academic honesty: As a student of University of Illinois at Chicago (UIC), you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty". Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Disability services: Concerning disabled students, the University of Illinois at Chicago is committed to maintaining a barrier-free environment so that individuals with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for full access and participation in UIC Programs must be registered with the Disability Resource Center (DRC). Center information can be found at http://www.uic.edu/depts/oar/campus_policies/disability_notification.html

Appropriate classroom behavior: The purpose of this is to ensure the equal right of learning.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.